

MAH Model Rebuild: Main-Text Model Candidate

Post-Phase-18 financing revision, September 2026

Abstract

This candidate is the restrained main-text representation of the current financing-adjusted derivation. The MAH reform makes retained entrusted manufacturing legally available by changing one institutional wedge. Developers choose one common project-advancement intensity before project characteristics and organizational routes are realized. Route choice is deterministic, and the price of qualified manufacturing services is endogenous. The baseline adds predetermined commercialization financing capacity only as a route feasibility margin. MAH changes organizational technology rather than finance supply. The baseline contains neither patent-generating research nor a fixed ranking between original and incremental project responses.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined research capability $a_i > 0$, internal manufacturing capability $k_i > 0$, and commercialization financing capacity $\ell_i > 0$, distributed according to the joint distribution $H(a, k, \ell)$. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q, m) = \sum_{g \in \{O, \text{Inc}\}} \rho_g F_g(q, m), \quad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \quad \tau_E(1) = \bar{\tau}_E < +\infty, \quad \bar{\tau}_E \geq 0. \quad (1)$$

The available routes are internal manufacturing I , retained entrusted manufacturing E , non-retained transfer T , and abandonment or delay A . Under E , the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as original-drug innovation investment / project-advancement intensity. It is broader than pure clinical development, but excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of viable projects reaching route planning is

$$\lambda_i^{\text{plan}} = a_i x_i. \quad (2)$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price. Second, after observing (a_i, k_i, ℓ_i, M) , they choose x_i . Third, planning-stage

projects draw (q, m) , check route financing requirements, and choose a route. Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. Upstream research and patent generation may help determine the predetermined a_i , but they are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p; q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c$$

and the gross operating present value

$$R(q, c) = \frac{Aq}{1 - \beta\varphi} \frac{(\varepsilon - 1)^{\varepsilon-1}}{\varepsilon^\varepsilon} c^{1-\varepsilon}, \quad \beta \in (0, 1), \quad \varphi \in [0, 1). \quad (3)$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$. Its route value is

$$W_i^I(q, m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \quad (4)$$

It requires a liquidity commitment $J_I(m, k_i) > 0$, with $J_{I,m} > 0$ and $J_{I,k} < 0$, and is financially feasible if and only if $J_I(m, k_i) \leq \ell_i$.

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m ,

$$W_i^E(q, m; M, p_m) = s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) - \mu_E - \tau_E(M). \quad (5)$$

It requires $J_E(m) > 0$, with $J'_E(m) > 0$, and is financially feasible if and only if $J_E(m) \leq \ell_i$. The J 's are liquidity requirements, not additional costs. Each monetary cost enters once. The transfer and abandonment values are

$$W^T(q, m) = T(q, m), \quad W^A = 0. \quad (6)$$

Define $\widetilde{W}_i^I = W_i^I$ when $J_I \leq \ell_i$ and $-\infty$ otherwise, and define $\widetilde{W}_i^E = W_i^E$ when $J_E \leq \ell_i$ and $-\infty$ otherwise. MAH changes none of ℓ_i, J_I, J_E .

3.3 Organization and project advancement

At a fixed p_m , the project chooses deterministically:

$$\begin{aligned} \widetilde{W}_i(q, m; M, p_m, \ell_i) &= \max\{\widetilde{W}_i^I, \widetilde{W}_i^E, W^T, W^A\}, \\ r_i^*(q, m; M, p_m, \ell_i) &\in \arg \max_{r \in \{I, E, T, A\}} \widetilde{W}_i^r. \end{aligned} \quad (7)$$

Continuous heterogeneity makes ties a zero-measure event; no logit or inclusive value is used.

The internal-minus-entrusted gap is

$$\begin{aligned} \Delta_{IE}(k_i) = & s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ & + F_E(m) + p_m b(m) + \mu_E + \tau_E(M). \end{aligned} \quad (8)$$

Holding (q, m, M, p_m) fixed conditional on both retained routes being financeable,

$$\Delta_{IE,k} = s(q)R_c(q, c_I) c_{I,k} - F_{I,k} > 0.$$

When the endpoint crossing conditions hold on the common-financeability set, a unique conditional cutoff $k^*(q, m; p_m, M)$ satisfies $\Delta_{IE}(k^*) = 0$. Conditional on I and E dominating T and A , developers below the cutoff select E , while those above it select I . Transfer and abandonment remain valid outside options.

Expected optimized value per planning-stage project is

$$\Omega_i(M, p_m) = \int \widetilde{W}_i(q, m; M, p_m, \ell_i) dF(q, m). \quad (9)$$

Developer i solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1 + \nu} x_i^{1+\nu} \right\}, \quad \kappa > 0, \quad \nu > 0. \quad (10)$$

Strict concavity gives the unique solution

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}, \quad (11)$$

including $x_i^* = 0$ when $\Omega_i = 0$. The binary reform affects x_i^* only through Ω_i . At a fixed p_m , its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j , with efficiency z_j , chooses capacity $s_j \geq 0$:

$$\max_{s_j \geq 0} \{p_m s_j - \Psi(s_j; z_j)\}. \quad (12)$$

With $\Psi_{ss} > 0$, positive-price capacity is uniquely determined by $p_m = \Psi_s(s_j^*; z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m, z) dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m; M, \ell_i) = \int b(m) \mathbf{1}\{r_i^*(q, m; M, p_m, \ell_i) = E\} dF(q, m). \quad (13)$$

Study-related and total demand are

$$\begin{aligned} D_m^{\text{MAH}}(p_m; M) &= \int a_i x_i^*(M, p_m) \chi_i^E(p_m; M, \ell_i) dH(a, k, \ell), \\ D_m(p_m; M) &= D_m^B(p_m) + D_m^{\text{MAH}}(p_m; M). \end{aligned} \quad (14)$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. Continuous heterogeneity makes aggregate demand continuous even though individual indicators can jump.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \quad (15)$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

The baseline partial equilibrium contains exactly

$$\{p_m^*, x_i^*, r_i^*(q, m), s_j^*\}_{i,j}. \quad (16)$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Main predictions

Organizational sorting. Conditional on I and E dominating the outside options, low internal manufacturing capability selects entrusted production and high capability selects internal production. Lower entrusted-route friction expands the entrusted region; a higher fixed CMO price contracts it.

MAH-relevant projects and advancement. The reform has a zero effect whenever E is not financeable or its value does not exceed the best financeable old route. Under the local ordering $J_E < J_I$, the interval $J_E \leq \ell_i < J_I$ is the financing corridor in which retained outsourcing is financeable but internalization is not. The reform response need not be globally monotone in ℓ_i , because higher financing also restores internalization as an old option. MAH does not change a_i , ℓ_i , project quality, or the advancement-cost function.

CMO equilibrium and scarcity. The qualified-capacity price is unique. Reform-induced entrusted demand can raise this price, weakly reducing the direct fixed-price gains in project value and advancement. Fixed-price comparisons and equilibrium-price comparisons are therefore distinct.

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF(q, m). \quad (17)$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \text{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.